

GridPowerAgent: A Grid-Aware LLM Agent for Power System Event Understanding and Tool-Orchestrated Decision Support

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Abstract—Large language models (LLMs) could act as a coordination layer between grid information, analysis tools, and operators—provided they read operating states accurately, select appropriate tools, and avoid hallucinated advice. We present GridPowerAgent, an agent that observes simulated power-system states, retrieves operating procedures, and orchestrates power-flow, contingency, and optimal-power-flow tools, evaluated on a seeded, fully regenerable corpus: 16,000 operating points and 15,000 scenarios across the IEEE 14/39/118 systems, ten disturbance classes each carrying power-flow ground truth, noisy measurements, state estimates, and rule-based labels. The pilot exposed a flaw in the benchmark itself: on the original key, both models scored zero on the two outcome-labeled classes (E6 undervoltage, E8 thermal overload) while describing the injecting cause mechanism correctly in nearly every failed response; the corpus now repairs this mixed-axis labeling by keying every scenario to its injected mechanism. Re-measured on the revised taxonomy, diagnosis reaches 100% (API) and 96.8% (local)—the pre-revision parity was an artifact, with all 18 vs 0 paired discordances favoring the API tier ($p=7.6e-06$). An observation ablation separates what diagnosis measures: from EMS-grade telemetry alone the API tier still achieves 93.9%, a local probe drops to 43.8% ($p=9.1e-13$), and from raw summary state diagnosis collapses (7.9%)—telemetry-based diagnosis is the realistic and tier-discriminative test. Tool selection retains a style difference (the API model defaults to power flow in 57% of picks), and hallucinated rows remain at most 0.54% of responses. All results carry exact denominators; findings are observations about this corpus at pilot scale, with a powered evaluation as the next step.

Index Terms—power system operation, large language models, LLM agent, tool orchestration, retrieval-augmented generation, state estimation

I. INTRODUCTION

Power-system operators are surrounded by analytical tools—power flow, state estimation, contingency analysis, optimal power flow—yet the work of interpreting their outputs, relating them to operating procedures, and deciding what to do next remains manual and expertise-bound. Large language models (LLMs) are candidates for an *intelligent coordination layer* in this workflow: not replacing conventional analysis, but reading grid states, retrieving the right procedures, invoking the right tools, and explaining the results [1], [2]. Recent agentic systems move in this direction [3], [4], but published evaluations rarely state what the labels are, where they come from, or how the scoring could be reproduced.

This paper presents GridPowerAgent and its evaluation under five research questions: (RQ1) how accurately can an LLM identify normal, abnormal, and critical operating

conditions; (RQ2) can it determine which engineering tool a given event requires; (RQ3) can it construct valid tool calls; (RQ4) does grid-specific retrieval improve operational reasoning; and (RQ5) does tool grounding reduce hallucination. RQ3—constructing valid tool calls—is answered here at its execution level: stated tools are executed against the scenario’s post-event network after the run and their execution scored (Sec. VII-F). This pilot answers RQ1, RQ2, RQ4, and RQ5, and RQ3 at the tool-name/execution level; argument-level validity is deferred to the powered sweep. The pilot also probes what diagnosis measures, via three prompt-side observation conditions (event description, EMS-grade telemetry, raw state; Sec. VII-C). We answer them on a seeded, headless-resumable corpus pipeline: 16,000 operating points and 15,000 scenarios across IEEE 14/39/118—ten disturbance classes (E0–E9) each carrying pre/post power-flow truth, noisy measurements, state estimates, severity, and rule-based reference labels—plus a FAISS knowledge base of operating procedures and four validated physics tools; configurations (E1 LLM-only, E2 +RAG, E3 +Tools, E4 Full) instantiate the ablation.

The evaluation compares two deployment tiers under identical paired prompts—a 4-bit-quantized small model served locally, and a lightweight API model—so conclusions do not depend on one provider. All figures carry exact denominators and Wilson CIs; power is stated alongside every null result; and one finding receives particular attention: both models systematically fail on two outcome-labeled disturbance classes, for a reason traced to the label taxonomy itself rather than to either model.

Contributions: (i) a grid-aware agent that couples grid-state observation, procedure retrieval, and validated physics tools; (ii) a seeded, hashed, fully regenerable 16k/15k/9.5M-measurement corpus with rule-based tool supervision and a quantified label-noise bound under estimation uncertainty; (iii) a dual-definition tool-scoring protocol that exposes and removes a degenerate answer strategy; (iv) a paired, exactly-denominated pilot across two deployment tiers in which a mixed-axis labeling flaw was identified, quantified, and repaired—turning an apparent diagnosis parity into a significant tier gap—together with an observation ablation separating description-reading from telemetry-based diagnosis; and (v) a per-check validation disclosure—including unresolved IEEE-39 islanding cases with their scenario identifiers shipped alongside the corpus—together with a seeded IEEE-39 replication of the full paired protocol (Sec. VII-D).

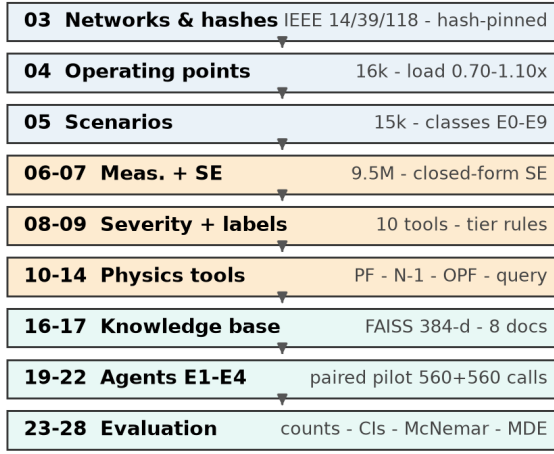


Fig. 1. Pipeline (Stages 03–28).

II. RELATED WORK

LLMs have been explored for power-system analysis assistance, dispatch, and contingency response [1], [2], including agentic orchestrators [3], [4] and dispatch benchmarks [5], but published evaluations seldom disclose how reference labels are produced; our corpus couples per-scenario power-flow truth, measurements, state estimates, and rule-based tool supervision across three IEEE sizes under one seeded pipeline, scored with exact denominators. Foundational RAG [6] and tool-augmented LLMs [7], [8], [9] motivate the agent design; state estimation and measurement modeling follow classical formulations [10], [11]; contingency (N-1) and AC-OPF are standard [12], [13], [14]; calibration is measured via ECE [15]; hallucination is judged against taxonomies [16]. Synthetic scenarios inherit the spirit of grid-ML benchmarks [17], [18]; pandapower [19] supplies AC power flow; FAISS [20] with sentence-transformers [21] supplies retrieval. On quantization: 4-bit precision is near-optimal for inference scaling [22] and standard post-training practice [23], though it measurably degrades small models—a conservative bias in our local deployment (Sec. VIII).

III. METHODOLOGY

Fig. 1 overviews the pipeline. Stages 03–05 build networks, operating points, and scenarios; 06–09 synthesize measurements, state estimates, severity, and reference labels; 10–14 expose physics tools; 16–17 build the knowledge base; 19–22 run the agent ablations; 23–28 evaluate and render figures. Heavy stages are idempotent and headless-resumable, materializing the full corpus in ~ 30 min.

Agent workflow. The agent follows an observe–diagnose–retrieve–plan–execute–interpret loop: it receives the post-event structured grid state (voltage magnitudes, branch loadings, outages, storage state of charge); optionally receives the top- k retrieved operating procedures (E2/E4) and a tool manifest (E3/E4); and must return the disturbance class, a tool selection, and a recommendation with confidence, as JSON. The harness scores the response against the rule-based reference labels; it

TABLE I
CORPUS. VALIDATION IS REPORTED PER SYSTEM, NOT AS A SINGLE GLOBAL CERTIFICATE.

Case	OPs	Scen. (per class)	Meas.	Auto checks	Worst V
14 (ref)	4k	3k (300)	361k	21/21	0.89 pu
39	5k	5k (500)	1.50M	20/21	0.89 pu
118	7k	7k (700)	7.68M	21/21	0.89 pu
Total	16k	15k	9.54M	—	—

IEEE-14 limits are artificial (3%/4%) to expose E8. The single failing check (s5_no_nan on IEEE-39) covers 41 scenarios with NaN post-voltages from two islanding outages; identifiers and causes ship with the corpus (case39_nan_scenarios.csv).

does not execute tools inside the agent loop in this pilot—stated tools are executed post hoc for an execution-validity check (Sec. VII-F). The diagnosis prompt presents the revised cause-axis taxonomy (Sec. VII-A), and three observation conditions (description, telemetry, state-only; Secs. VI and VII-C) probe what the diagnosis measures.

Networks and scenarios. Networks use pandapower IEEE 14/39/118 cases with tuned thermal limits to make compound events observable (14: 3% line / 4% transformer; 118: 6%; 39: nameplate). Limits are disclosed per system and deliberately *not* comparable across systems (Sec. VIII). Operating points sweep load 0.70–1.10 $\times$ with $\pm 5\%$ bus-local noise, renewable fractions, and storage state of charge 0.15–0.85 (20 MW/40 MWh at bus 9 on IEEE-14). Topology hashes pin all three networks. Ten classes are injected: E0 Normal; E1 load surge, E2 load drop, E3 line outage, E4 generator outage, E5 renewable ramp (cause classes); E6 undervoltage, E7 overvoltage, E8 thermal overload (outcome classes, physically iterated to target severity); E9 compound. Reference labels key every scenario to its injected mechanism class; the original outcome-class keying of E6/E8, the flaw it created, and its repair are analyzed in Sec. VII-A.

IV. CORPUS, SIMULATION AND VALIDATION

Table I summarizes the corpus. Scenarios inject the ten classes with 300/500/700 per class on 14/39/118; replay over 40 draws confirms exact determinism.

Measurements model $\sigma_V = 0.003$ pu and $\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05)$ MVA computed from the noise-free power-flow solution (Eqs. 1–2). State estimates are a closed-form noise-aware approximation ($\hat{v} = v + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.0007^2)$), *not* full iterative WLS with assembled Jacobians; replacement with `pandapower.estimation.estimate` is future work and we do not claim paper-grade estimator fidelity.

$$z = h(x) + \epsilon, \quad \epsilon_V \sim \mathcal{N}(0, 0.003^2) \quad (1)$$

$$\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05), \quad W = \text{diag}(\sigma^{-2}) \quad (2)$$

$$\hat{x} = \arg \min_x (z - h(x))^T W (z - h(x)) \quad (3)$$

TABLE II
DIAGNOSIS UNDER THE REVISED TAXONOMY, EXACT COUNTS WITH WILSON 95% CIs.

Cfg	API ($n=140$)	Local ($n=140$)
E1	140/140 [97, 100]	137/140 [94, 99]
E2	140/140 [97, 100]	133/140 [90, 98]
E3	140/140 [97, 100]	138/140 [95, 100]
E4	140/140 [97, 100]	134/140 [91, 98]

TABLE III
PAIRED DIAGNOSIS ANALYSIS (LOCAL — API, PERCENTAGE POINTS) UNDER THE REVISED TAXONOMY. BOOTSTRAP 95% CI, 20K RESAMPLES; ALL DISCORDANCES ARE ONE-DIRECTIONAL.

Cfg	Pairs	Diff (pp)	95% CI	McNemar p
E1	140	-2.1	[-5.0, 0.0]	0.25
E2	140	-5.0	[-8.6, -1.4]	0.02
E3	140	-1.4	[-3.6, 0.0]	0.50
E4	140	-4.3	[-7.9, -1.4]	0.03

from the revised key, and presents the cause-axis taxonomy to the agent. Scenario identifiers, post-event physics, and the outcome flags are unchanged; E7 overvoltage is kept, since its injection mechanisms match the outcome reading. The revised-key pilot (Secs. VII-C–VII-D) uses the same 140 scenarios, paired prompts, and scoring pipeline.

B. Re-Measured Diagnosis on the Revised Taxonomy (RQ1/RQ4)

Table II reports exact counts under the revised key. The API model diagnoses 140/140 in *every* configuration; the local model diagnoses 133–138/140 (95.0–98.6%). All 18 vs 0 paired discordances favor the API model (pooled exact McNemar $p=7.6e-06$; individually significant in E2 and E4)—the statistical parity observed under the flawed key (77.9% vs. 76.2%, exact McNemar not significant) was an artifact of the mixed axis, not evidence of equivalent capability. Configuration additions move local diagnosis by at most five scenarios in either direction while the API tier stays at ceiling: no diagnosis benefit from retrieval or tool manifests is detectable at this scale, which answers RQ4 negatively *for diagnosis*; the minimum detectable difference is E1: 3.4, E2: 5.2, E3: 2.8, E4: 4.8 pp.

C. Observation Ablation: Reading the Log vs. Reading the Grid

The near-ceiling diagnosis above is measured with a plain-language description of the injected event in the prompt. Three observation conditions (Sec. VI) separate what diagnosis measures on the API tier: with the description, 100.0%; with the description replaced by EMS-grade telemetry (switching status, changed demand and renewable elements, system-wide pre→post deltas), 93.9%; with the raw summary state alone, 8.0%. The state-only collapse is degenerate—the model answers E7 for 95% of scenarios—because the summary state carries essentially no cause information. The telemetry

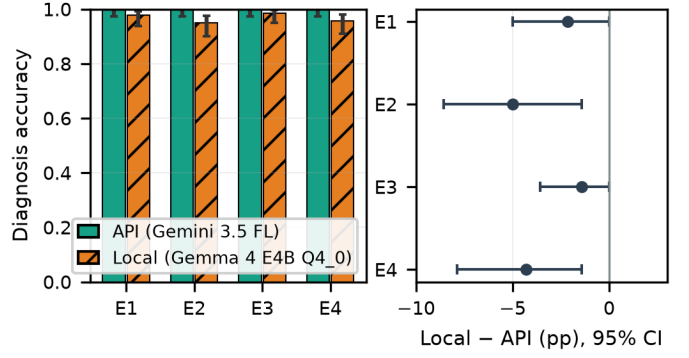


Fig. 3. (a) Diagnosis accuracy under the revised taxonomy, local vs. API model, Wilson 95% CIs. (b) Paired diagnosis difference (Local — API) with bootstrap 95% CIs.

condition is the realistic and discriminative one: the switch-detectable classes (E3, E4, E7) are diagnosed perfectly and compounds reach 94.1%, with one systematic failure—E0 normal operation is called E7 in 0/20 of cases. The telemetry condition also separates the deployment tiers: on a 20-scenario probe (80 cells), the local model reaches 43.8% against 95.0% for the API tier (41 vs. 0 discordant, $p=9.1e-13$), with hedged multi-class answers and JSON contract breaks among its errors. Diagnosis from telemetry rather than description is therefore both the realistic deployment condition and the discriminative test.

D. Cross-System Check: IEEE-39 (RQ7: Cross-System Transfer, Begun)

A seeded 140-scenario draw from the IEEE-39 corpus (excluding the 41 islanding-NaN scenarios) runs the same paired protocol under the pre-revision prompt; its logged answers are rescored against the revised key. The API model diagnoses 140/140 scenarios in *every* configuration (E1–E4) and the local model 135–138/140. Strict tool selection declines monotonically for the API model—57, 52, 43, and 31 of 140 across E1–E4—while the local model stays flat at 62–63/140. The decline extends the IEEE-14 pattern (58, 56, 49, and 43 of 140 there) but is steeper and does not plateau by E4, while the local model is flat on both corpora: accumulating context and tool manifests appears to progressively disrupt the API tier’s strict tool selection—flagged for the powered sweep. Transferring between corpora changes the network topology, load patterns, and label instances while the agent, prompt, and scoring pipeline remain untouched—the strongest test this pilot size permits.

E. RQ2: Tool Selection and the Default-Answer Bias

Tool selection is where the deployment tiers differ most—in *style* before *accuracy*. The permissive accepted-set metric is degenerate (always answering power flow achieves it by construction; random choice scores 53%), and the API model exercises exactly that strategy: 57% of its stated tools are power flow. Under the strict-specific metric, the local model names a scenario-*required* tool in 75/140–82/140 of cases (flat across configurations) versus 43/140–58/140 for

TABLE IV

TOOL SELECTION UNDER THE STRICT-SPECIFIC METRIC (DEFINED IN THE EVALUATION PROTOCOL), EXACT COUNTS WITH WILSON 95% CIs.

Cfg	API ($n=140$)	Local ($n=140$)
E1	58/140 [34, 50]	81/140 [50, 66]
E2	56/140 [32, 48]	77/140 [47, 63]
E3	49/140 [28, 43]	75/140 [45, 62]
E4	43/140 [24, 39]	82/140 [50, 66]

the API model. The conditioning behind these rates sharpens the default-answer-bias reading. When the API model states power flow (317 responses), that tool is actually required for the scenario in only 41% of cases—the rest are defaults to the universally accepted answer. The local model states power flow rarely (73 responses, 26% of them required); its dominant choices are contingency (315 responses, 100% required—contingency is required precisely on outage and overload scenarios) and equipment queries. The gap therefore reflects the API model’s default rate, not superior local tool knowledge; both models remain far from ceiling, and the binary metric cannot distinguish well-placed tools from lucky guesses.

F. RQ3: Tool-Call Execution Validity

The harness executes every stated tool on the scenario’s post-event network and scores the call. Execution is a property of the tool and scenario, not the model, so pairs are pooled across models and configurations: 537 stated-tool pairs, of which 521 are executable (the remainder specified no tool) and every executable call succeeds—power flow converges on every reconstructed network, contingency executions converge and reproduce the scenario’s recorded overload, N-1 sweeps complete all line outages, and OPF solves. RQ3 is therefore answered affirmatively at the tool-name/execution level: every stated tool executes without error. Execution is, however, only a necessary condition for the valid tool-call construction RQ3 asks about: whether arguments are correct and the call targets the right contingency element is not checked. Argument-level validity (component identifiers, subcommands) requires an extended prompt schema and remains deferred to the powered sweep.

G. RQ5: Hallucination and the Cost of Local Inference

Any-tag hallucinated rows are rare for both models under the automated judge: 0/0/0/0 (E1–E4) (API) and 2/0/0/1 (E1–E4) (Local) across E1–E4. These rates mean “the rule-based judge flagged no row”; they are not expert-annotated ground truth. Confidence calibration is unidentifiable at this sample size and will be reported on the powered sweep. Latency is the deployment trade-off: local inference averages 44–49 s per call versus 1.06–1.11 s for the API round-trip—roughly a 44 $\times$ difference—while carrying zero marginal API cost, no rate-limit dependency, and no grid data leaving the premises; acceptable for always-on monitoring, but neither model’s latency profile suits closed-loop use (Sec. VIII).

VIII. DISCUSSION AND LIMITATIONS

Scope. The evidence covers *one model pair* (a 4-bit small local model; a lightweight API tier) on *one corpus*, one prompt template, temperature 0, single runs. Results do not speak to frontier API models or claim that local models match API models in general. The minimum detectable difference at this sample size is $\sim 3\text{pp}$ (Sec. VI); the powered 600-scenario sweep is the inferential step.

Label circularity. Disturbances, labels, and prompts derive from the same rule family: the evaluation measures *recovery of the synthetic labeling policy*, not open-ended operator reasoning—and the severity boundaries of Eq. (4) were themselves tuned on this corpus, though the strict-metric comparison is invariant across four boundary sets, two of them not derived from this corpus (Sec. IV). Required remedies, all future work: expert-reviewed labels (2×80 , $\kappa \geq 0.8$), multiple valid tool sequences, blind scoring, an evaluator independent of the label generator, and externally anchored severity boundaries.

Taxonomy design. The E6/E8 label-axis failure (Sec. VII-A) was a benchmark-design finding: outcome-labeled classes injected via cause mechanisms made cause-correct answers score zero. The revision repairs it by rekeying E6/E8 to their injected mechanisms; both label versions ship with the corpus, the outcome information remains in the physics columns, and expert review of the revised labels plus externally anchored severity boundaries remain future work. The observation ablation adds a second encoding finding: the state-only view carries essentially no cause information (7.9% diagnosis, Sec. VII-C), and the API tier misreads every E0 normal scenario as overvoltage under telemetry-only observation.

Physics fidelity. Thermal limits differ per system by construction, so overload magnitudes are not operationally comparable across systems. The corpus is static: no dynamics, protection, or time-sequential behavior. The severity index is illustrative with a quantified label-noise bound (Sec. IV); the state estimator is a closed-form approximation pending replacement with pandapower’s iterative estimator; the 41 islanding-NaN IEEE-39 scenarios (identifiers shipped in `case39_nan_scenarios.csv`) remain unresolved and are excluded from downstream post-voltage use.

Scoring subjectivity. Hallucination tags come from a single automated rule-based judge (Sec. VII-F); tool scoring relies on the reference policy, and the strict metric narrows but does not eliminate policy-dependence.

Safety framing. This is an offline advisory prototype: no protection coordination, authorization, or human-approval interface is implemented; closed-loop use is out of scope.

Reproducibility and generalization. All generation scripts, seeds, hashes, prompt templates, validation summaries, the IEEE-39 NaN identifier list, both label versions, and the revised-taxonomy and observation-ablation run datasets are in the repository; corpora regenerate deterministically; an archived (DOI) release is planned. Generalization: the IEEE-39 cross-system evaluation (Sec. VII) is performed for both

deployment tiers; IEEE-118 transfer and a powered cross-system protocol remain future work, alongside the powered 600-scenario $\times$ 4-configuration sweep per model with per-class F1 under both label axes, ECE, multiplicity-corrected paired tests, and prompt/temperature sensitivity.

IX. CONCLUSION

GridPowerAgent couples a seeded, regenerable 16k/15k power-system scenario corpus with a grid-aware LLM agent that retrieves procedures and orchestrates validated physics tools, evaluated under a protocol where every number carries an exact denominator. In the paired pilot, a systematic diagnosis failure was traced to the benchmark’s own mixed-axis label taxonomy and repaired by rekeying the affected classes to their injected mechanisms; on the revised taxonomy the API tier diagnoses at ceiling (100%) and significantly above the local tier (96.8%), diagnosis from EMS-grade telemetry alone remains at 93.9% for the API tier while the local tier drops to 43.8%, and tool choice still differs mainly in default-answer bias. The agent, the corpus, and the protocol—including their disclosed flaws—are the contribution; the powered evaluation they now enable is the next step.

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